



Physical AI LEGO Quality Inspection Robot

- SmolVLA + ACT • Imitation Learning • H100 • Rerun.io
- Train a robot to identify, sort, and place LEGO blocks based on quality.

The Problem

Manual LEGO quality inspection and sorting is repetitive.

A robot needs to understand what it sees AND learn precise physical actions.

Goal: automatically classify each block as Perfect or Defective and place it correctly.

Our Approach

Human Leader Arm → Physical AI Data Collection

SmolVLA → Vision + language understanding

ACT → Learn robot movements from demonstrations

H100 → Imitation-learning training

Robot → Pick & place into Perfect / Defective plate

Learning From a Leader

A human/operator guides the leader arm through the task.

Robot demonstrations provide real-world state/action data.

The model learns the relationship between visual observations and robot actions.

This enables the follower robot to reproduce the learned manipulation behavior.

Debug & Replay with Rerun.io

Visualize demonstrations and robot trajectories.

Replay actions step by step.

Inspect observations, predictions, and robot behavior.

Use debugging data to identify failures and improve training.

Demo: From Vision to Action

1. Camera observes LEGO block
2. Model identifies quality: Perfect / Defective
3. ACT predicts the next robot action sequence
4. Robot grasps the block
5. Robot places it on the corresponding plate
6. Rerun.io records and visualizes the process

What We Demonstrate

Physical AI connecting perception, learning, and action.

Foundation-model-based robot learning with SmolVLA.

Imitation learning with ACT and real demonstrations.

GPU-accelerated training on an H100.

A practical debug/replay workflow for robot learning.